CulturalRiddles: A Multicultural Benchmark of Riddles

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Motivation

- Problem: Most cultural benchmarks test knowledge, but cultural understanding requires inferring implicit meanings from context.
- Why Riddles: A riddle packs language, history and local knowledge into one short question



You cannot understand “the German 9/11”
unless you know in the German context, it refers to 9 November.

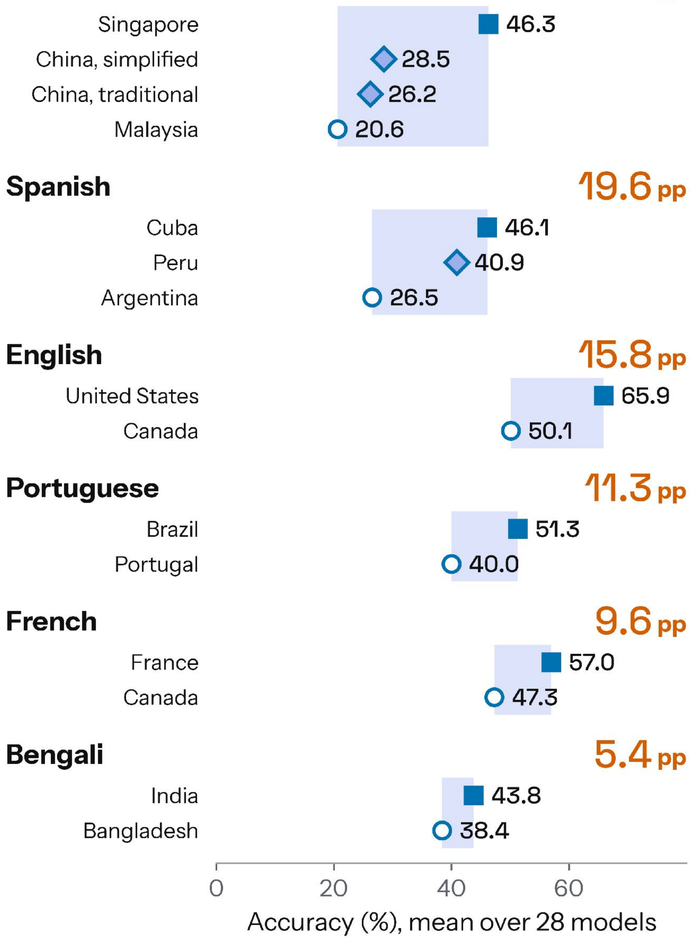
CulturalRiddles

6,191 original riddles written by **167 native speakers**, verified by community members



Each community favors its own riddle style

5–26 pp
gap between communities that share a language



98% of Luganda riddles close with “Who am I?”

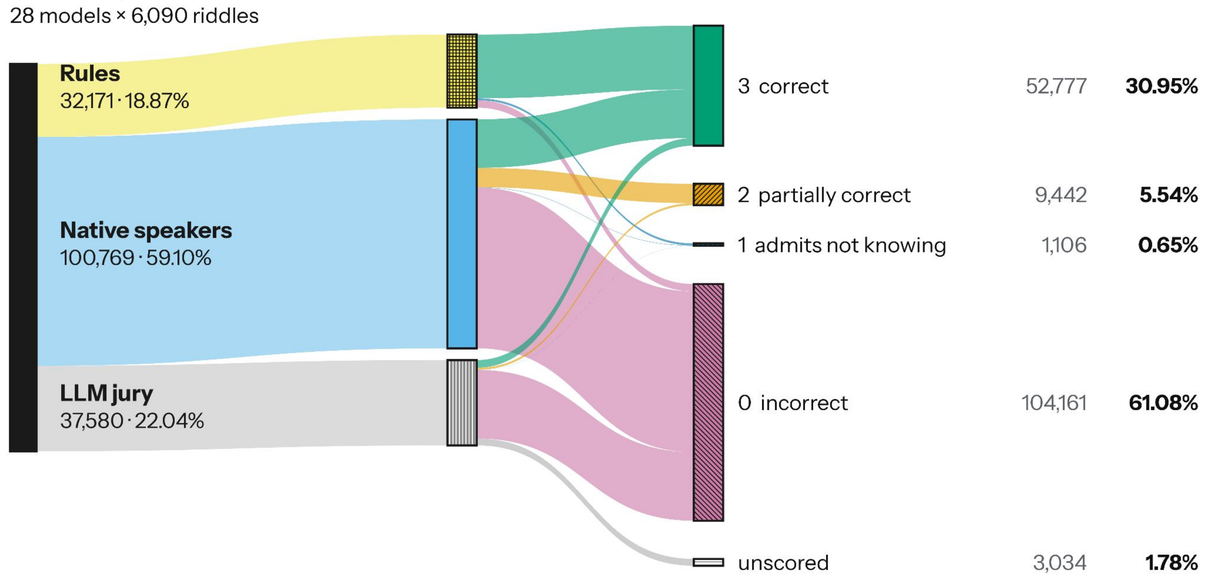
98% of Japanese riddles close with a “What is it?”

Communities sharing a language differ by 5–26pp

Evaluation

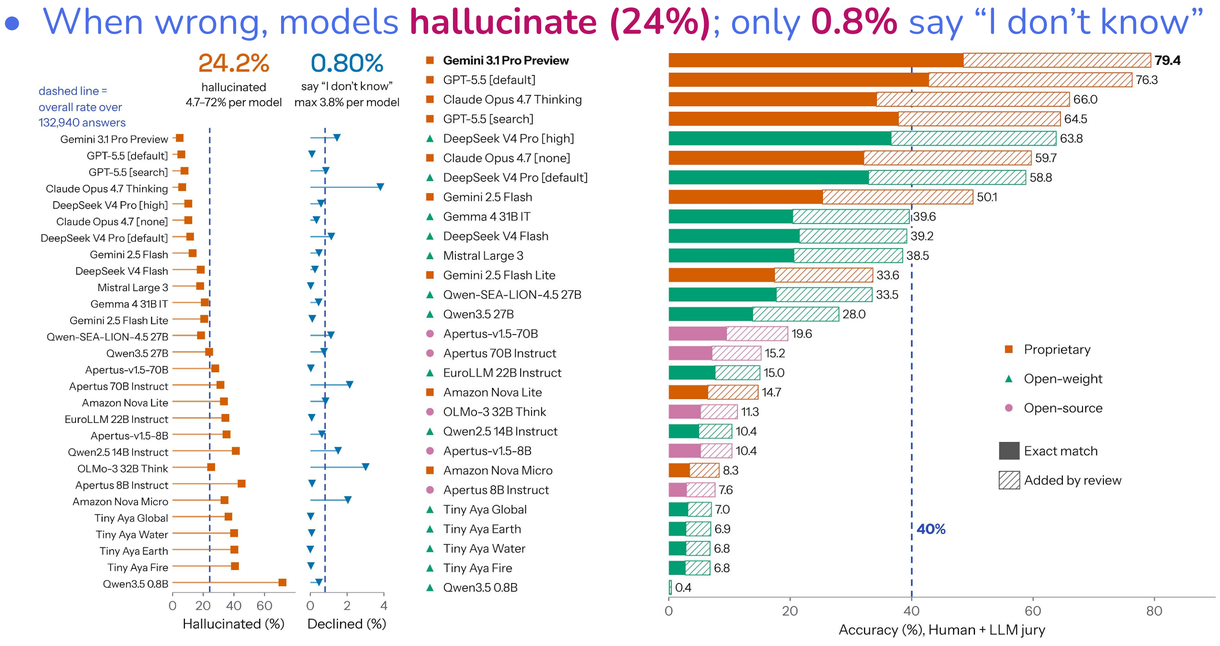
Native speakers scored 59% (100,769) of answers;
One of the **biggest** ever human annotation in NLP research 🎉

170,520
model responses
28 models × 6,090 riddles

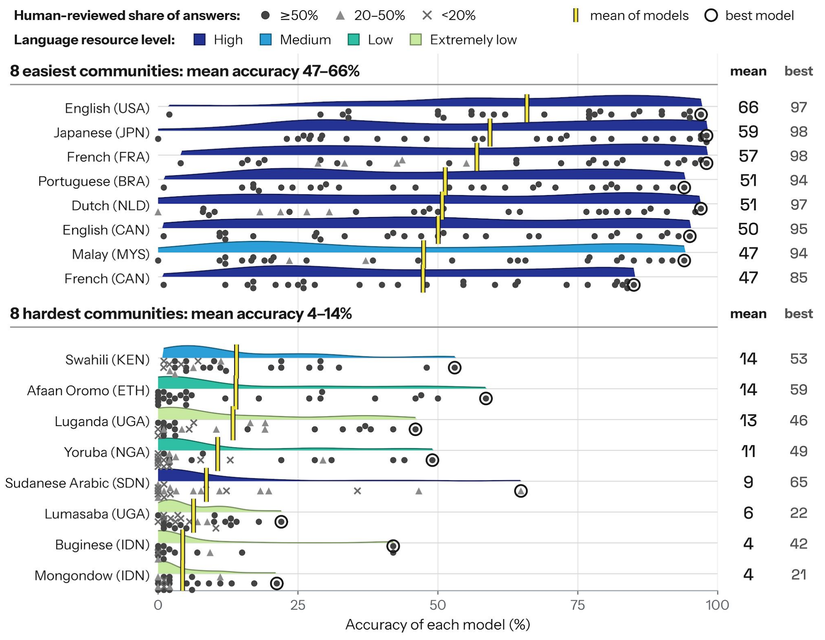


Main Findings

- When wrong, models **hallucinate (24%)**; only **0.8%** say “I don’t know”



- **High** averages hide cultural **blind spots**



The best model scores 79.4% overall, but all models score ≤22% on *Lumasaba* and *Mongondow*.

- Resource and Community Gaps Persist Despite English Translation Gains

